

3. An insurance company sells single premium deferred annuity contracts with return linked to a stock index, the time- t value of one unit of which is denoted by $S(t)$. The contracts offer a minimum guarantee return rate of $g\%$. At time 0, a single premium of amount π is paid by the policyholder, and $\pi \times y\%$ is deducted by the insurance company. Thus, at the contract maturity date, T , the insurance company will pay the policyholder

$$\pi \times (1 - y\%) \times \text{Max}[S(T)/S(0), (1 + g\%)^T].$$



You are given the following information:

- (i) The contract will mature in one year. $T=1$ ω
- (ii) The minimum guarantee rate of return, $g\%$, is 3%. ω $g=0.03$
- (iii) Dividends are incorporated in the stock index. That is, the stock index is constructed with all stock dividends reinvested. **NO DIVIDENDS!**
- (iv) $S(0) = 100$.
- (v) The price of a one-year European put option, with strike price of \$103, on the stock index is \$15.21.

Determine $y\%$, so that the insurance company does not make or lose money on this contract.

The Synopsis:

- (A) 12.8%.
- (B) 13.0%
- (C) 13.2%
- (D) 13.4%
- (E) 13.6%.

① Focus on the insurance company's liability $\star$

② Use our data to simplify $\star$

③ Algebraically manipulate $\star$
w/ the intent to "create"
the expression for the payoff of the put

→ The Insurance Company's Liability.

$$\pi (1-y) \max \left(\frac{S(T)}{S(0)}, (1+g)^T \right)$$

const

$$\frac{1}{S(0)} \max (S(T), (1+g)^T \cdot S(0))$$

$$\frac{(1.03) \cdot 100}{103}$$

$$\max (S(T), 103)$$

$$V_p(T) = (K - S(T))_+$$

$$\max(S(T), 103) = ?$$

a, b

$$\begin{aligned}\max(a, b) &= a + \max(0, b-a) = a + (b-a)_+ \\ &= b + \max(a-b, 0) = b + (a-b)_+\end{aligned}$$

$$\max(S(T), 103) = \boxed{S(T)} + \boxed{(103 - S(T))_+}$$

Long
Stock

Payoff of the put w/ strike 103
and exercise date $T=1$.

The insurance company can perfectly hedge by:

- Longing/Buying $\frac{\bar{\pi}(1-y)}{S(0)}$ units of the stock index;
- Buying $\frac{\bar{\pi}(1-y)}{S(0)}$ European puts w/ $K=103$ and $T=1$.

If they receive the same amount of money @ time 0 as is the cost of this replicating portfolio, they BREAK EVEN.

$$\cancel{\bar{\pi}} = \frac{\cancel{\bar{\pi}}(1-y)}{S(0)} (S(0) + V_p(0))$$

$$100 = (1-y)(100 + 15.21)$$

$$1-y = \frac{100}{115.21} \Rightarrow y = \frac{15.21}{115.21} = \underline{13.2\%}$$

